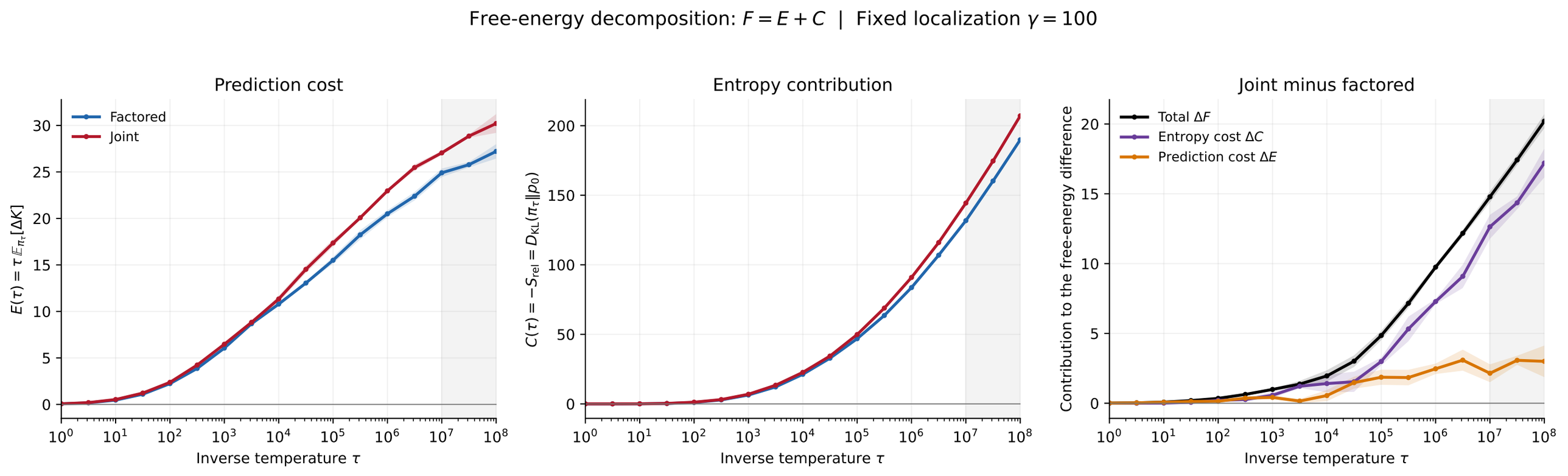
Free-energy decomposition: $F = E + C$ | Fixed localization $\gamma = 100$
Prediction cost
Factored
Joint
$E(\tau) = \tau\, \mathbb{E}_{\pi_\tau}[\Delta K]$
Inverse temperature τ
Entropy contribution
$C(\tau) = -S_{\mathrm{rel}} = D_{\mathrm{KL}}(\pi_\tau \| p_0)$
Inverse temperature τ
Joint minus factored
Total ΔF
Entropy cost ΔC
Prediction cost ΔE
Contribution to the free-energy difference
Inverse temperature τ
Longer HMC, 128 particles, four runs per arm. Bands: ±2 Monte Carlo SE with covariance. Shading: upper-range sensitivity.